Before the Lab Acts: Benchmarking Language Models for Constrained Compound Selection

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Abstract

Language models are increasingly connected to scientific tools and automated laboratories, but many evaluations stop at predictions, prose, or unconstrained plans. We introduce SpecGuard-Chem, an action-level benchmark for the point at which assay evidence becomes an experimental commitment. A *decision card* is one self-contained assay-level task containing 50 measured support compounds, a fixed candidate pool, explicit eligibility rules, and a budget of 10 selections. We include all 91 CARA lead-optimization tasks that can support this top-10 action, rather than selecting the largest pools. Candidate outcomes are physically separated into a hash-bound scorer artifact. The primary outcome, *feasible utility*, is the sum of hidden pChEMBL values recovered by valid selected candidates; invalid or missing positions receive zero benchmark credit. A prespecified matrix compares basic and descriptor-enriched interfaces for three dated provider conditions. A *guarded* pipeline checks a parsed LLM shortlist and deterministically fills invalid or missing positions; it is a model-plus-software system, not the LLM alone. The strongest guarded pipeline achieved mean feasible utility 73.964 and normalized discounted cumulative gain at 10 (NDCG@10) 0.929, but the best assay-local quantitative structure–activity relationship (QSAR) model remained higher by 1.003 mean card-level utility points (paired 95% CI 0.405–1.647). The same pre-repair response stream had mean feasible utility 72.966 and issued a fully valid action on 72 of 91 cards. Repair was invoked on 19 cards but supplied only 14 of 910 final positions in that best condition; the Anthropic and DeepSeek conditions depended much more heavily on fallback. SpecGuard-Chem does not claim a new biological dataset or a complete model of drug discovery. It contributes a reproducible action-evaluation layer over CARA for testing whether an LLM adds value before it is trusted to spend laboratory resources.

Keywords: AI for science; automated laboratories; compound prioritization; lead optimization; large language models; CARA; QSAR; benchmarking.

1 Introduction

Automated laboratories couple computational experiment selection to physical execution. Robot-scientist and self-driving-laboratory systems have already integrated model-guided screening, chemical optimization, and robotic experimentation [18, 4]. More recently, LLM-based systems have planned chemical tasks, called expert tools, and controlled laboratory interfaces [2, 3]. As the distance between generated text and physical action narrows, evaluation must ask more than whether an answer sounds plausible: would the exact issued action be executable, and is it a good use of a finite experimental budget?

LLMs have been studied in molecular science through several distinct routes. Mol-Instructions developed domain-specific instruction data spanning molecular, protein, and biomolecular-text tasks [6]. ChemLLMBench and ChemBench assessed chemical knowledge and reasoning across heterogeneous tasks and showed substantial dependence on task and evaluation setting [7, 11]. LLM4SD used knowledge extracted or inferred by an LLM as features for conventional molecular-property models [20], whereas ChemCrow used an LLM to coordinate specialist chemistry tools [3]. These results motivate hybrid scientific systems, but they do not establish whether direct LLM selection adds value over a model trained on assay-local measurements.

This paper isolates one decision primitive from a future drug-discovery loop: choosing the next assay batch from a known pool using sparse project-local activity evidence. The boundary is deliberately narrow. It excludes compound generation, synthesis planning, multi-assay optimization, and sequential learning. In return, every available action, constraint, model input, outcome, and opportunity cost can be audited exactly, and LLM-backed systems can be compared with transparent molecular rankers on identical information.

The activity-data substrate is CARA, which organizes experimentally measured compound activities into assay-local virtual-screening and lead-optimization tasks [15, 16]. CARA found that models trained separately on the support examples for an assay were strong comparators in its few-shot lead-optimization setting. At the inference level, our task is candidly CARA-style activity estimation plus eligibility filtering and top-k allocation.

We do not claim a new raw dataset or an unrelated prediction problem. The contribution is the action-level protocol: each system must issue exactly k executable candidate identifiers, raw and repaired behavior remain separate, and LLM pipelines are compared against per-card fingerprint regressors on identical visible evidence.

We ask three questions:

1. **Decision utility:** do LLM-backed selectors issue higher-utility assay batches than random-valid, heuristic, similarity, and strong per-assay QSAR systems?
2. **Interface and representation:** does descriptor-enriched molecular evidence change shortlist utility relative to a basic interface?
3. **Executability and operational value:** how often are raw LLM actions valid, what does deterministic repair change when applied to the same response, and what latency/token/dollar cost buys any gain?

Decision utility is the paper's primary scientific question. Validity is an independent property: a compliant weak list is executable but unhelpful, while a promising invalid list cannot be run as issued. The benchmark therefore does not treat guardrail gains as scientific gains.

2 Positioning: from prediction to action

Compound-activity prediction benchmarks typically score per-compound estimates or rankings. CARA improves the realism of that evaluation by retaining assay-based tasks, sparse labels, and lead-optimization versus virtual-screening settings [15]. SpecGuard-Chem retains that substrate and adds an action contract and systems protocol at the point where predictions would be converted into a laboratory allocation.

Table 1: CARA is the data and task substrate; SpecGuard-Chem is an action-evaluation layer. The distinction is the evaluated output and its executable semantics, not a claim of unrelated molecular learning.

	CARA activity view	**SpecGuard-Chem action view**
System output	Query activity predictions or ranking	Ordered batch of exactly k candidate IDs
Evaluation unit	Compound/task predictive performance	Whole experimental allocation per task
Budget	Not intrinsic to each prediction	Explicit finite assay capacity
Constraints	Molecular prediction domain	Candidate eligibility plus executable output contract
Failure semantics	Prediction error	Opportunity cost and/or unexecutable action
Artifact protocol	Activity benchmark data/splits	Separate system inputs and hash-bound scorer outcomes; exact requests, complete traces, costs, repair attribution

We call each self-contained assay-level selection problem a decision card. SpecGuard-Chem converts CARA prediction tasks into cards with a finite budget, an output contract, and explicit failure semantics. It further reports scientific shortlist quality separately from executability and attributes any deterministic repair. This differs from instruction-tuning, broad chemistry question answering, property-prediction, and tool-coordination studies. The comparator ladder is therefore part of the contribution: an LLM result is meaningful only relative to small, reproducible rankers trained on the same support evidence.

3 Benchmark construction

3.1 Corrected CARA import

The release uses CARA v1.0.1 and the official `LO_All` table and split [16]. Split JSON files associate each task key with integer row positions. The corrected importer resolves each integer positionally, then verifies that the source row's task identifier equals the split key. Out-of-range positions, task mismatches, duplicate candidate identities, and support–candidate identity overlap are hard errors.

An exhaustive audit resolves all 24,588 references: 5,000 support and 19,588 query rows across 100 tasks. The previous label-based importer was incorrect; all artifacts from the earlier 50-card draft and their downstream results are retired rather than treated as an earlier benchmark version.

3.2 Decision cards and biological context

For each eligible task, the public artifact contains the CARA task, target, and endpoint identifiers; 50 support structures with measured activity; every candidate ID, structure, and permitted descriptor; constraints; and budget $k = 10$. Candidate activity is stored only in a separate evaluator artifact. Each outcome row is bound to the canonical hash of its public card, preventing labels from being silently paired with a changed task.

The measured quantity is CARA's supplied *pChEMBL Value*, a standardized negative-log molar activity scale on which higher values indicate greater potency. Every card and request states this scale and direction explicitly; endpoint labels such as IC50, Ki, EC50, and Potency retain the assay context but do not change the higher-is-better interpretation of the supplied pChEMBL value.

All 100 official `LO_All` tasks were considered. The sole inclusion rule was whether a card contained at least ten feasible candidates, because an exact top-10 action is otherwise impossible. Ninety-one tasks met that rule. The nine excluded tasks had feasible counts 0, 0, 0, 0, 0, 0, 0, 6, and 8; every exclusion was recorded before model evaluation, and hidden outcomes were never used for inclusion. No largest-pool or outcome-based subsampling was performed. This removes candidate-pool size as a task-selection criterion, but it does not make the benchmark representative of CARA virtual-screening tasks, the nine infeasible lead-optimization tasks, or industrial projects.

Table 2: Frozen v0.1.0 composition. Each task has a distinct CARA target identifier.

Property	**Value**
Official CARA tasks considered	100
Included / excluded tasks	91 / 9
Support compounds per task	50
Total public candidates	18,205
Candidate-pool range / mean	52–967 / 200.055
Feasible-pool range / mean	12–579 / 110.165
Endpoint tasks	68 IC50; 17 Ki; 5 EC50; 1 Potency
Assay budget	10

Target and endpoint identifiers provide minimal biological context and enable task-coherence checks. They do not supply mechanism, selectivity, phenotypic context, ADMET, toxicity, or a multi-objective project profile. This is an assay-local potency decision benchmark, not a broad biological-reasoning test.

3.3 Action contract and constraints

For task c, let E_c be the visible support evidence, P_c the candidate pool, $V_c \subseteq P_c$ the candidates satisfying molecular eligibility, and $k = 10$. A system m emits a schema-conforming action for task c containing an ordered sequence $S_m(c)$ of candidate IDs. The action is valid exactly when the task identity matches, it has length k, it contains unique members of P_c, it excludes support identities, and every selected candidate lies in V_c.

Version 0.1.0 candidate eligibility requires an RDKit-parseable structure, molecular weight at most 500 Da, and RDKit Crippen cLogP at most 4.5. The 500-Da limit follows the familiar rule-of-five upper bound [10]. The prespecified 4.5 cLogP cutoff is a simplified benchmark threshold, 0.5 below the conventional rule-of-five value of 5.0; it was fixed before model evaluation. Across all 19,588 official candidate records, every structure parsed, 10,039 passed both limits, 2,433 exceeded molecular weight only, 3,765 exceeded cLogP only, and 3,351 exceeded both. The configured forbidden-substructure list was empty. These thresholds provide transparent, reproducible, non-trivial constraints for testing specification adherence. They are benchmark design choices, not definitions of drug-likeness, oral bioavailability, synthesis feasibility, medicinal-chemistry suitability, or safety.

Wider future loop (not evaluated)

design $\longrightarrow$ make $\longrightarrow$ test $\longrightarrow$ analyse $\longrightarrow$ update

SpecGuard-Chem action boundary

support evidence + finite pool $\longrightarrow$ filter, estimate, rank $\longrightarrow$ ordered top-10 assay batch
constraints + budget $\longrightarrow$ system under test $\longrightarrow$ hidden retrospective outcomes

Figure 1: The benchmark isolates one transition at which a model-generated ranking becomes a finite experimental commitment. The surrounding autonomous laboratory loop is motivation, not an evaluated claim.

4 Systems and interfaces

4.1 Non-language comparators

We include a random-valid selector; a descriptor-desirability heuristic; similarity to the most active support compound using Morgan fingerprints [14]; and random-forest, gradient-boosting, and linear support-vector regressors trained separately for each card on only its 50 visible support measurements [13, 17]. Structures are represented by 2,048-bit radius-2 Morgan fingerprints. Random forest uses 100 trees, minimum leaf size 1, and seed 7. Gradient boosting uses the locked scikit-learn defaults and seed 7. Linear SVR uses $C = 1.0$, a linear kernel, and sparse-compatible standard scaling without centering. No card-specific hyperparameter optimization is performed. Each model predicts all feasible candidates, which are sorted by predicted activity with candidate ID as a deterministic tie-break. A non-deployable oracle ranks feasible candidates using hidden outcomes and defines attainable headroom. Parsing, descriptors, and fingerprints use RDKit [9]; exact software versions are recorded in the supplement and lockfile.

4.2 LLM interfaces and frozen conditions

The final matrix crosses 91 cards, two interfaces, and three model conditions (546 raw provider requests). Each request contains one complete card: task and assay identifiers; the output contract; the 50 support IDs, structures, and measured pChEMBL values; the ordered candidate IDs and structures; constraints; and budget. The *basic* interface also supplies candidate molecular weight and cLogP. The *descriptor-enriched* interface adds precomputed TPSA, hydrogen-bond donor/acceptor counts, and rotatable-bond count. Hidden candidate activities and external search are absent from both interfaces. Candidate ordering and prompt wording are fixed within each condition.

The frozen conditions are OpenAI `gpt-5.5-2026-04-23` with low reasoning effort, Anthropic Claude Opus 4.8 without extended thinking, and `deepseek-v4-pro` with thinking disabled, as selected and checked on 16 July 2026 [12, 1, 5]. Every condition allows at most 4,096 output tokens. Temperature is not set and therefore follows the provider default; no provider generation seed is set. OpenAI and DeepSeek use provider JSON-object response modes. Anthropic uses an explicit JSON-only prompt followed by deterministic extraction and validation of one JSON object. SDK retries are disabled, and the matrix contains exactly one successful recorded provider attempt per request. Thus "raw" below means the parsed pre-repair output of that single response, not necessarily a provider-native structured object.

Exact requests were frozen before execution. Configured and returned model identifiers, response identifiers, settings, token counts, latency, finish reasons, and raw content are retained. A conservative estimate placed the full uncached matrix at $106.06, with an explicit $120 stop gate. The authorized execution completed with 58.957 dollars of usage-derived token-pricing cost. No call occurs unless the command also receives `-allow-external`; exact replay uses the committed cache.

4.3 Post-hoc deterministic repair

A *guarded LLM system* is the LLM plus deterministic software that checks and, when needed, repairs its action. Here "guarded" refers only to compliance with the benchmark contract; it does not imply biological, chemical, or clinical safety. Raw and guarded behavior are evaluated from the same provider response. The validator checks JSON shape, task identity, selection count, candidate membership, duplicates, support exclusion, and candidate constraints; it never reads activity outcomes. Valid, unique raw IDs are retained in serialized response order, and unoccupied positions are filled from the fixed rules-only ranking before revalidation. Schema or rank correction can therefore count as a repaired card without replacing a candidate identity. The raw response, raw issues, repaired output, and repair attribution remain stored. This design avoids conflating a different repair prompt or an additional paid model call with deterministic harness behavior. The outcome-blind fallback formula is given in the supplement; it computes TPSA even for the basic interface, so guarded results include software-only descriptor access that the corresponding raw LLM did not receive.

5 Evaluation

5.1 Utility, ranking, and regret

Let s_{mcr} be the candidate ID returned by system m for card c at position r, with a missing symbol when that position was not supplied. Define a_{mcr} as $y_{s_{mcr},c}$ only when the position exists, its ID is a feasible candidate with an available hidden outcome, and that ID has not occurred at an earlier scored position; otherwise set $a_{mcr} = 0$. Only the first k returned positions are scored. *Feasible utility* is therefore

$$U_m(c) = \sum_{r=1}^{k} a_{mcr}. \tag{1}$$

Thus the first occurrence of an otherwise valid ID can score, whereas later duplicates, out-of-pool or infeasible IDs, missing outcomes, missing positions, and positions beyond k receive no credit. Zero is an executability penalty, not an imputed activity. Because every card requires ten positions, $U_m(c)/10$ is the mean pChEMBL recovered per required slot.

All source labels share the negative-log molar pChEMBL scale and higher-is-better direction, but IC50, Ki, EC50, Potency, assay conditions, and activity distributions are not biologically interchangeable. The primary comparison is therefore paired within card. For systems A and B, we first calculate $\Delta_c = U_A(c) - U_B(c)$ on the same assay, support evidence, and candidate pool, then equally weight the 91 card-level differences. For valid fixed-size actions, an assay-specific additive offset cancels from this contrast. Aggregate utility is a benchmark summary, not a clinical or cross-assay physical quantity.

NDCG@k is a secondary, within-card normalized measure of ordered ranking quality relative to that card's ideal feasible top-10 ordering [8]; 1.0 is ideal. Constrained regret is

$$R_m(c) = \sum_{i \in O_c} y_{ic} - U_m(c), \tag{2}$$

where O_c is the feasible hidden-outcome top-k. Lower regret is better.

5.2 Action validity and operational outcomes

Whole-action validity is a binary card-level measure: it equals one only when the complete output has no contract issue, including schema shape, task identity, exact size, in-pool membership, uniqueness, support exclusion, and candidate eligibility. We separately report the valid-selection fraction (`compliance_rate`), defined as valid selected entries divided by k, as a partial-credit diagnostic rather than a synonym for action validity. For example, ten feasible selections returned under the wrong task identifier have selection compliance one but action validity zero. For every LLM interface we report raw metrics and the corresponding post-hoc-repaired metrics, plus repair rate, fallback-supplied positions, and violation taxonomy. Latency, input/output tokens, cache status, and estimated or actual cost are reported per condition.

5.3 Bootstrap analysis

Every resampling unit is a complete decision card. Marginal 95% confidence intervals for a system mean use 1,000 bootstrap samples from NumPy's default random-number generator with seed 7. Each sample draws 91 cards with replacement; its mean is calculated, and the 2.5th and 97.5th percentiles form the interval. Headline system contrasts use a paired bootstrap: the 91 matched card-level differences are calculated first, then resampled with replacement 2,000 times using seed 13. The observed mean difference and the corresponding percentile interval are reported in the prespecified direction, including best-QSAR minus best-guarded-LLM. Pairing preserves shared card difficulty; bootstrapping estimates uncertainty but does not make assay endpoints biologically equivalent or correct task selection, dependence beyond the card, or model-selection multiplicity. Because the displayed best systems were identified on this benchmark, their intervals are descriptive post-selection intervals, not multiplicity-adjusted confirmatory tests or Bayesian posterior probabilities.

6 Results

6.1 The corrected benchmark separates molecular selectors

Table 3 establishes the benchmark's useful dynamic range before provider spending. Eligibility filtering alone is not sufficient: rules-only utility is 66.922 and random-valid utility is 68.469. Similarity rises to 73.288, while all three assay-local QSAR models reach approximately 74.75–74.97. The oracle remains at 79.563, leaving 4.596 utility points of mean headroom above the best deployable baseline. All rows have perfect action validity, so their separation is attributable to ranking rather than malformed actions.

Table 3: Corrected deterministic results on all 91 cards. Utility intervals are task-level 95% bootstrap percentile intervals; NDCG@10 is a point estimate. Oracle is not a deployable system.

System	Mean utility (95% CI)	NDCG@10	Regret	Action validity
Oracle valid top-k	79.563 (77.486–81.812)	1.000	0.000	1.000
QSAR linear SVR	74.966 (72.830–77.032)	0.938	4.596	1.000
QSAR random forest	74.958 (72.825–77.142)	0.938	4.605	1.000
QSAR gradient boosting	74.750 (72.621–76.859)	0.935	4.813	1.000
Similarity to best active	73.288 (71.264–75.236)	0.919	6.274	1.000
Random valid	68.469 (66.487–70.450)	0.855	11.094	1.000
Rules/desirability	66.922 (64.772–69.062)	0.828	12.641	1.000

6.2 Do LLMs add decision value?

The strongest final LLM pipeline was the OpenAI basic interface plus deterministic post-hoc repair, with mean feasible utility 73.964 and NDCG@10 0.929. Linear-SVR QSAR remained higher by 1.003 mean card-level utility points (paired 95% CI 0.405–1.647). The corresponding QSAR-minus-guarded-LLM NDCG@10 difference was 0.0087 (95% CI 0.0010–0.0165). These two outcomes support a QSAR-family lead rather than added decision value from the tested LLM pipelines.

The underlying pre-repair OpenAI response stream had mean feasible utility 72.966 and issued a fully valid action on 72 of 91 cards (0.791). Guarding raised utility to 73.964, but this is not the performance of the LLM alone. Repair was invoked on 19 cards; only ten cards received a replacement identity, and deterministic fallback supplied 14 of 910 final positions (1.54%). Nine repairs corrected metadata, schema, or ranking details without replacing an identity.

Harness contribution was much larger elsewhere (Table 4). Fallback supplied 25.16–27.69% of Anthropic final positions and 39.01–58.68% of DeepSeek positions. Thirty DeepSeek basic-interface cards received all ten final identities from fallback. Those guarded scores primarily characterize a model-plus-deterministic-harness system. Descriptor enrichment did not improve the best-performing OpenAI condition, and effects varied by provider rather than supporting a general descriptor benefit. The six raw conditions cost 58.957 dollars in total under provider-reported usage and frozen token-pricing rules; the best condition accounted for 11.623 dollars. Repair added no provider calls.

Table 4: Pre-repair output and deterministic fallback contribution. A repaired card can have zero replaced identities when only schema, task metadata, or rank normalization changed. Final guarded rows are model-plus-harness results.

Provider, interface	Cards repaired	Cards with identity replacement	Fallback positions	All 10 from fallback
OpenAI, basic	19/91 (20.88%)	10/91	14/910 (1.54%)	0
OpenAI, descriptors	23/91 (25.27%)	10/91	15/910 (1.65%)	0
Anthropic, basic	74/91 (81.32%)	71/91	252/910 (27.69%)	1
Anthropic, descriptors	69/91 (75.82%)	68/91	229/910 (25.16%)	1
DeepSeek, basic	83/91 (91.21%)	81/91	534/910 (58.68%)	30
DeepSeek, descriptors	69/91 (75.82%)	69/91	355/910 (39.01%)	12

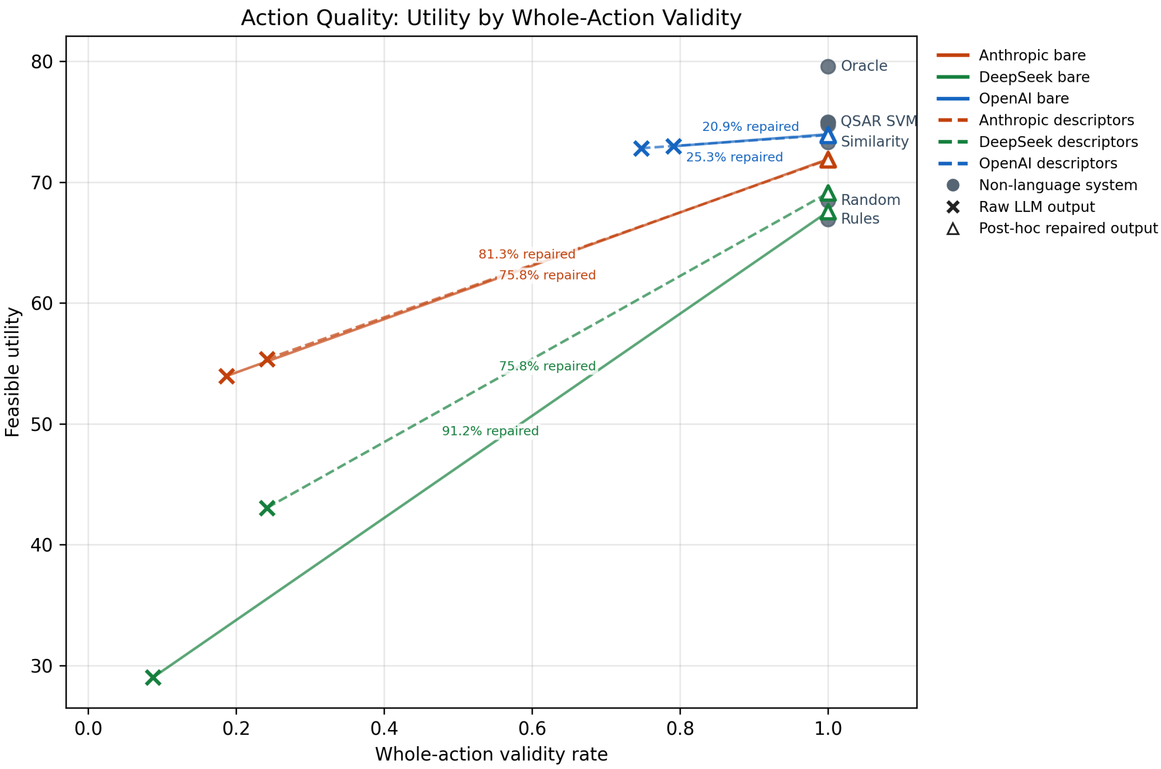

Figure 2: Action utility versus whole-action validity. Crosses are raw LLM outputs; open triangles are deterministic post-hoc-repaired outputs from the same responses. Connecting lines show harness intervention, not an additional model call; repaired points must not be interpreted as model-only performance.

7 Discussion

7.1 A benchmark with a deliberately simple core

The formal core of this task is filter–predict–rank–allocate. This simplicity does not establish a new molecular-learning problem, and we do not claim otherwise. It does create a precise evaluation boundary at the point where a model's scores become an experimental commitment. That boundary makes invalid actions, opportunity cost, deterministic repair, system inputs, and operational cost measurable together.

The corrected evidence shows that the benchmark has independent legs: constraints do not collapse the task, transparent molecular models improve materially on heuristic selection, and oracle headroom remains. The tested LLMs did not beat the best assay-local QSAR systems. For this bounded action, a decision system should therefore retain the smaller specialized ranker as the strong default and treat an LLM, if used, as an auditable component rather than assuming it is the scientific decision engine.

7.2 Relationship to existing evidence

The strength of per-card QSAR is consistent with CARA's lead-optimization analysis, where models trained on assay-relevant examples performed strongly and performance varied among assays [15]. Variation among providers and interfaces is likewise consistent with chemistry benchmarks showing that LLM performance depends on task and evaluation setup [7, 11]. Mol-Instructions, LLM4SD, and ChemCrow demonstrate useful roles for domain instruction data, LLM-derived features, and tool coordination [6, 20, 3]; none implies that a directly prompted LLM should outperform a regressor fitted to local assay measurements. Our result is narrower: on this corrected retrospective top-10 benchmark, guarded pipelines produced executable shortlists, but the strongest conventional QSAR retained a paired advantage.

7.3 What the benchmark does not establish

This is a retrospective computational benchmark, not prospective lead optimization. It includes every one of the 91 `LO_All` tasks that can support the declared top-10 action, but it does not represent CARA virtual-screening tasks, the nine tasks with fewer than ten feasible candidates, or the distribution of industrial projects. CARA also inherits ChEMBL measurement and curation heterogeneity [19]. Bootstrap intervals describe variation among these observed cards and do not license generalization beyond them.

The objective is assay-local potency, not selectivity, mechanism, uncertainty, ADMET, toxicity, pharmacokinetics, synthesis, novelty, availability, or therapeutic value. The candidate pool is fixed and no structure is generated. The molecular thresholds are simplified eligibility rules; no structural alert was active, and no medicinal chemist comparator was included. Consequently, "feasible" means feasible under the benchmark's rules, not suitable for synthesis or development.

Each provider condition retains one response per card, so repeated-call variation is not estimated. Results are specific to the dated models, prompts, interfaces, parsing, settings, and deterministic fallback. CARA and ChEMBL are public: candidate outcomes were withheld at prompt time and from non-oracle ranking code, but prior model exposure to public source data cannot be excluded. One-shot allocation also does not test learning across rounds. These limits preclude claims about prospective laboratory performance, autonomous-laboratory readiness, or LLMs in drug discovery generally.

8 Reproducibility and release

The frozen working release separates public cards from scorer outcomes and includes source and transform provenance, exclusions, exact requests, model configurations, traces, per-card scores, aggregate tables, costs, environment locks, licenses, citation metadata, and reproduction commands. Historical invalid results remain immutable but are explicitly superseded. A semantic release tag and the final top-level manifest/checksum freeze remain separate release-administration actions.

9 Conclusion

Across all 91 eligible CARA lead-optimization cards, the strongest assay-specific QSAR model outperformed the strongest tested guarded LLM pipeline on the paired primary outcome. Deterministic validation made final actions executable, but fallback supplied between 1.54% and 58.68% of final shortlist positions across conditions. Guarded performance must therefore be reported as model-plus-harness performance alongside the pre-repair model result. This study supports strong conventional comparators, paired card-level analysis, and explicit repair attribution when evaluating constrained LLM decision systems; it does not establish superiority or general utility of LLMs for prospective drug discovery.

Data, code, and AI-use statement

Release artifacts and exact reproduction commands accompany the repository. Code is MIT licensed; CARA and CARA-derived data retain CC BY 4.0 terms. ChatGPT/Codex assisted with project framing, code and artifact auditing, manuscript structuring/editing, and LaTeX compilation. All numerical claims are checked against local machine-readable artifacts. No patient or clinical data are used. Author list, contributions, funding, competing interests, and venue-specific disclosure wording require human confirmation before submission.

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